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Extensions of a classical mechanics “piston-model” for understanding the impact of asymmetry on ICF implosions: The cases of mode 2, mode 2/1 coupling, time-dependent asymmetry, and the relationship to coast-time

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ABSTRACT

As long suspected, low mode asymmetry in inertially confined fusion (ICF) implosions has been implicated as a performance limiting factor [Casey *et al.*, “Evidence of three-dimensional asymmetries seeded by high-density carbon-ablator nonuniformity in experiments at the national ignition facility,” *Phys. Rev. Lett.* **126**, 025002 (2021)]. Recently a non-linear, but solvable, theory [Hurricane *et al.*, “An analytic asymmetric-piston model for the impact of mode-1 shell asymmetry on ICF implosions,” *Phys. Plasmas* **27**, 062704 (2020)] based upon the simple picture of a pair of asymmetric pistons has generated new insights and provided some practical formulas for estimating the degradation of an implosion due to mode-1 asymmetry and demonstrated a previously unrecognized connection between measured hot-spot drift velocity, nuclear down-scatter ratio asymmetry, and the concept of residual kinetic energy (RKE). Asymmetry of the implosion “shell,” as opposed to asymmetry of the hot-spot, was key to the classical mechanics model because the majority of the kinetic energy in an implosion is carried by the shell. Herein, the two-piston model is extended to a six-piston model in order to capture mode-2 asymmetry and coupling between mode-1 and mode-2. A key result of this new six-piston model is that the weighted harmonic mean of shell areal density is the fundamental quantity that determines the RKE and performance degradations for a three-dimensional implosion. Agreement is found between the scalings coming from the theory and ICF implosion data from the National Ignition Facility and to large ensembles of detailed simulations. The connection between the piston model’s dependence upon the radius of peak velocity and coast-time is also highlighted in this paper. Finally, by extending the two-piston model to include time-dependent “swing,” it is shown in the Appendix that the shell asymmetry at the time of stagnation dominates the solution for RKE.

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I. INTRODUCTION

Recently, a great deal of evidence has been amassed, which shows that inertial confinement fusion (ICF) implosions on the National Ignition Facility (NIF) have been tangibly impacted by asymmetry. The causes of the asymmetries have been identified as coming from non-uniformities in the shell thickness during shell manufacture,¹ laser beam power/energy imbalances,³ and x-ray drive non-uniformities from diagnostic window cutouts present in the hohlraum.⁴ A view-factor analysis has been successful in rolling-up the various causes of asymmetry and determining the direction of the net

hot-spot mode-1 drift velocity that appears consistent with trends seen in experiments for the vast majority of implosions.⁵

The performance impact of mode-1 asymmetry has been shown to be well modeled with a relatively simple two-piston model representation of an ICF implosion.² The two-piston model demonstrated that hot-spot metrics of performance, such as hot-spot pressure (p_{hs}), hot-spot energy (E_{hs}), hot-spot temperature (T_{hs}), and no-burn fusion field (Y_{noz}), all degrade as various powers of $(1 - f^2)$, where f , the shell asymmetry fraction, is a measure of shell areal density ($\rho\delta R$) asymmetry at stagnation. In particular, f was shown to be related to measurable

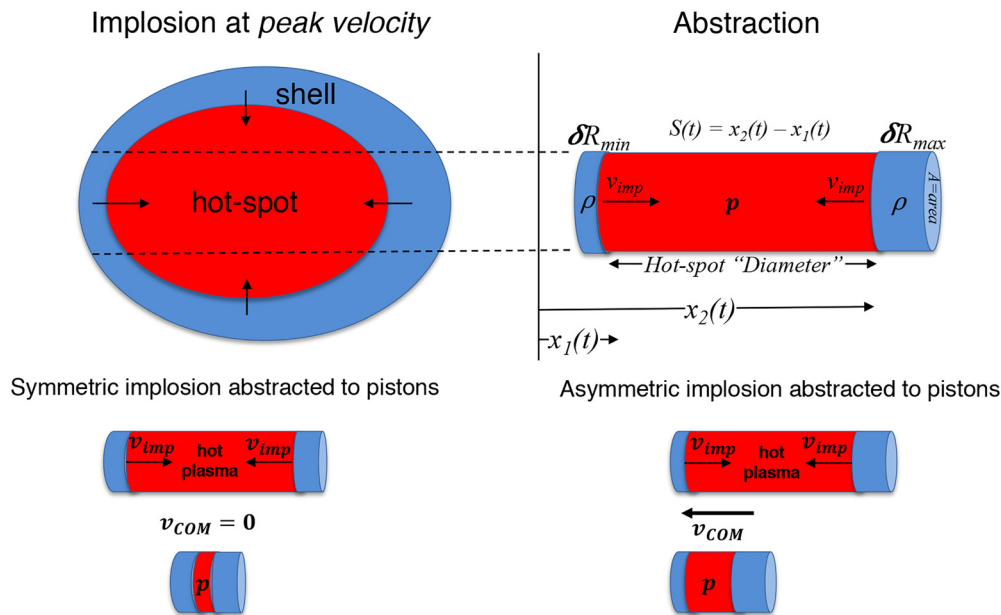


FIG. 1. (Upper left) An asymmetric implosion with a mode-1 distortion in the hot-spot and shell. (Upper right) An asymmetric piston model abstraction of an implosion useful for analysis of mode-1. (Lower left) With no mode-1 asymmetry the pistons converge symmetrically upon the hot-spot and there is no net center-of-mass (COM) motion and all of the piston kinetic energy is converted into hot-spot internal energy at stagnation. (Lower right) With a mode-1 asymmetry, the thin piston decelerates more rapidly than the thick piston, and there is a net COM motion toward the thin piston. The kinetic energy associated with the COM motion is never converted into internal energy of the hot-spot.

quantities of down-scatter ratio (*DSR*, assumed proportional to shell areal density^{6,7}) asymmetry at stagnation, hot-spot drift velocity (v_{hs}), and the concept of residual kinetic energy (*RKE*) that has been considered in past asymmetry studies.^{8–11} Namely,²

$$f \equiv \frac{\rho\delta R_{max} - \rho\delta R_{min}}{\rho\delta R_{max} + \rho\delta R_{min}} = \frac{v_{hs}}{v_{imp}} \approx \sqrt{\frac{RKE}{0.66 \cdot KE}}, \quad (1)$$

where v_{hs} is calculated from the time-integrated neutron spectra¹² along multiple lines-of-sight that allow one to determine the bulk burn-averaged fluid velocity in the fusing plasma.¹³ In the model that led to Eq. (1), the direction of the hot-spot velocity is determined by the shell center-of-mass motion.

The two-piston model highlighted that it is the asymmetry of the shell of an implosion which is important to the performance of an implosion and that focusing upon hot-spot asymmetry, while easy to diagnose, results in misleading conclusions. Physically, the importance of the shell can easily be understood by noting that the majority of an implosions kinetic energy and confinement come from the surrounding shell not the hot-spot, as Fig. 1 simply illustrates.

In this paper, by considering a higher dimensional extension of the piston model—a six-piston model (see Fig. 2)—and by considering a time-dependent version of the two-piston model, it will be shown that the shell asymmetry at stagnation is dominant asymmetry in terms of implosion performance and that the hot-spot asymmetry is a higher-order correction except in the case of extreme asymmetry. In particular, it will be shown that in general, a weighted harmonic mean (WHM) of shell $\rho\delta R$ is the fundamental measure of asymmetry that determines implosion performance and that Eq. (1) above is a special case of the WHM for pure mode-1 asymmetry. From the general

WHM solution, explicit practical RKE formulas will be provided for specific combinations of shell and hot-spot asymmetry and turning points where these solutions appear to break down will be pointed out. Finally, an interesting connection between coast-time¹⁴ and the radius of peak velocity that is part of the piston-model solution will be made explicit, showing why an implosion stagnation pressure increases with reduced coast-time as has been noted previously.

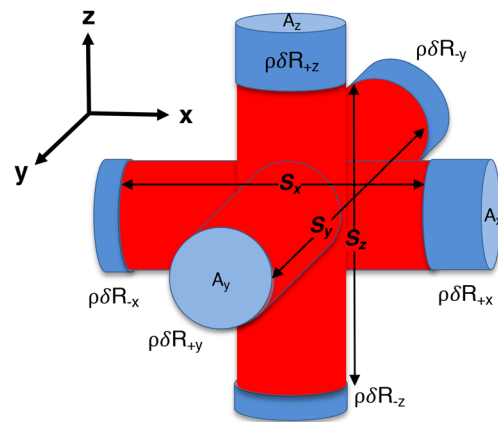


FIG. 2. Shown is an abstraction of 3D implosion that is made up of 6 pistons doing work on a common hot-spot. The configuration is that of three cylinders at right angles to each other. This geometry of the intersection volume is known as a “Steinmetz solid” and the calculation of the volume is non-trivial. The pairs of pistons that lay along a particular coordinate have a time-dependent separations, S_x , S_y , and S_z . Each piston has its own areal density and cross-sectional area.

II. SIX-PISTON MODEL FOR MODE-2 AND MODE-2/I COUPLING

In analogous to Ref. 2, an abstraction of a realistic three-dimensional (3D) implosion is constructed from an arrangement of pistons initially converging toward each other with velocity (v_{imp}). If initially the pistons have different speeds, the solution can be performed, without a loss of generality, in a frame of reference moving with the piston-group center-of-mass velocity.

Newton's law gives the deceleration of each piston, shown in Fig. 2, in response to the increasing pressure developing between the pistons as the pistons converge upon each other,

$$\begin{aligned}\ddot{x}_{\pm x} &= \frac{\pm P}{\rho \delta R_{\pm x}}, \\ \ddot{y}_{\pm y} &= \frac{\pm P}{\rho \delta R_{\pm y}}, \\ \ddot{z}_{\pm z} &= \frac{\pm P}{\rho \delta R_{\pm z}},\end{aligned}\quad (2)$$

where the $\rho \delta R$ terms are the areal density of each piston.

By differencing the piston equations of motion in Eq. (2) along the same coordinate direction, one obtains three differential equations for the piston separations,

$$\begin{aligned}\ddot{S}_x &= \left(\frac{1}{\rho \delta R_{+x}} + \frac{1}{\rho \delta R_{-x}} \right) P, \\ \ddot{S}_y &= \left(\frac{1}{\rho \delta R_{+y}} + \frac{1}{\rho \delta R_{-y}} \right) P, \\ \ddot{S}_z &= \left(\frac{1}{\rho \delta R_{+z}} + \frac{1}{\rho \delta R_{-z}} \right) P.\end{aligned}\quad (3)$$

Now noting that the second time derivative of volume, for the configuration of Fig. 2, is related to the piston separations of Eq. (3), $\ddot{V} = A_x \ddot{S}_x + A_y \ddot{S}_y + A_z \ddot{S}_z$, we obtain a single differential equation for hot-spot volume. Namely,

$$\ddot{V} = \frac{2P_{pv}(A_x + A_y + A_z)}{\rho \delta R_{WHM}} \left(\frac{V_{pv}}{V} \right)^{5/3}, \quad (4)$$

where as in Ref. 2, an adiabatic pressure law is justified for implosions with low levels of α -heating, so $P(t) = P_{pv}[V_{pv}/V(t)]^{5/3}$, where $P_{pv} = P(0)$ is the hot-spot pressure at peak velocity and $V_{pv} = V(0)$ is the hot-spot volume at peak velocity.¹⁵ In the construction of Eq. (4), it is found that all the piston areal densities combine into a single key parameter, the weighted harmonic mean areal density,

$$\rho \delta R_{WHM} = \frac{\sum_j A_j}{\sum_j \frac{A_j}{(\rho \delta R)_j}}, \quad (5)$$

where the index j runs over the six-pistons of Fig. 2. Equation (5) emphasizes the low areal density regions, which is clearly different than the average areal density

$$\rho \delta R_{ave} = \frac{\sum_j A_j (\rho \delta R)_j}{\sum_j A_j}, \quad (6)$$

which emphasizes the high areal density regions. A similar areal density harmonic mean has been found independently.¹⁶ As will be shown below, the area weighting is important especially for large implosion asymmetries. The discrete form of Eqs. (5) and (6) suggests that in the continuous limit, integrating over a hot-spot surface area is appropriate

$$\rho \delta R_{WHM} \rightarrow \frac{\int dA}{\int \frac{dA}{\rho \delta R}} \quad \text{and} \quad \rho \delta R_{AVE} \rightarrow \frac{\int dA \rho \delta R}{\int dA}. \quad (7)$$

In Eqs. (5)–(7), information about the hot-spot asymmetry is in the area weightings, while the shell symmetry information is contained in $\rho \delta R$.

Equation (4) is integrable because the equation is autonomous, so writing $\dot{V} = \dot{V} dV/dV$ and integrating both sides of Eq. (4) yields

$$\frac{1}{2} \dot{V}^2 + const. = -\frac{3}{2} P_{pv} V_{pv} \frac{\sum_j A_j}{\rho \delta R_{WHM}} \left(\frac{V_{pv}}{V} \right)^{2/3}. \quad (8)$$

Using the initial condition (i.e., the conditions at peak velocity) that $V = V_{pv}$ and $\dot{V} = -v_{imp} \sum_j A_j$ at $t=0$ determines the integration constant. Then with the integration constant determined, one can solve for the minimum volume (i.e., hot-spot volume), when $\dot{V} = 0$, obtaining

$$\begin{aligned}\frac{V_{pv}}{V_{min}} &= \left[1 + \frac{\rho \delta R_{WHM} v_{imp}^2 \sum_j A_j}{3 P_{pv} V_{pv}} \right]^{3/2} \\ &\approx \left[\frac{\rho \delta R_{WHM} v_{imp}^2 \sum_j A_j}{3 P_{pv} V_{pv}} \right]^{3/2},\end{aligned}\quad (9)$$

where the approximation is valid in the ICF limit where the shell (i.e., piston) kinetic energy is much greater than the hot-spot internal energy at peak velocity. As in the mode-1 piston model,² with the hot-spot volume change in-hand, Eq. (9), one can quickly write down expressions for the low-burn (defined by levels of yield amplification < 2) stagnation pressure, $P_{stag} = P_{pv}(V_{pv}/V_{min})^{5/3}$, hot-spot internal energy, $E_{hs} = (3/2)P_{stag} V_{min}$, etc. For example, the full expression for P_{stag} is

$$P_{stag} \approx \frac{v_{imp}^5}{P_{pv}^{3/2}} \left[\frac{\rho \delta R_{WHM} \sum_j A_j}{3 V_{pv}} \right]^{5/2}. \quad (10)$$

Note that there is an important subtlety about Eq. (10) and the implied scalings when compared to a real or simulated implosion with convergence (see Appendix C). Since P_{pv} and V_{pv} are not usually measured, it is sometimes useful to have an alternate form of Eq. (10), namely, using Eq. (9) to eliminate the factor of $P_{pv}^{3/2} V_{pv}^{5/2}$ one obtains a stagnation pressure expression in terms of the minimum (hot-spot) volume

$$P_{stag} = \rho \delta R_{WHM} v_{imp}^2 \left(\frac{\sum_j A_j}{3 V_{min}} \right). \quad (11)$$

Note that for a sphere of radius R , the surface area over volume is $3/R$, so the term in parenthesis in Eq. (11) is effectively $\sim 1/R_{\min}$. Equation (11) makes it simple to write down the hot-spot internal energy,

$$E_{hs} = \frac{1}{2} \left(\sum_j A_j \right) \rho \delta R_{WHM} v_{imp}^2. \quad (12)$$

In writing down the expressions, e.g., Eqs. (10)–(12) [or just by looking at Eq. (4)], one can recognize that all the mode-1 piston model results² translate to the six-piston model results just using the transformation $(1 - f^2) \rightarrow (\rho \delta R_{WHM})/(\rho \delta R_{ave})$. A complete set of relations is given in Table I. As shown in Appendix A, the ratio $(\rho \delta R_{WHM})/(\rho \delta R_{ave}) \leq 1$ regardless of the form of asymmetry.

We can now construct the generalized relation for RKE by writing down the kinetic energy of the six-piston arrangement shown in Fig. 2,

$$KE = \left(\frac{1}{2} \sum_j (\rho \delta R)_j A_j \right) v_{imp}^2. \quad (13)$$

Then using Eq. (12) with $RKE = KE - E_{hs}$ to find

$$\frac{RKE}{c \cdot KE} = 1 - \frac{\rho \delta R_{WHM}}{\rho \delta R_{ave}}, \quad (14)$$

where $c=1$ in the piston model abstraction (since the pistons are assumed incompressible), but an average value of $c \approx 0.66$ is found for compressible shells (by comparing to simulations as discussed in Ref. 2). When comparing to ICF data or simulations where the inflight adiabat ($\alpha_{if} > 1$) of the fuel is known a more precise expression, $c = 1/(1 + 1.53/\alpha_{if}^{1.35})$ can be used, noting that for higher adiabat implosions a larger fraction of shell KE is split into the hot spot than for low adiabat implosions.

While simple, Eq. (14) is extremely powerful and it says that RKE is simply related the WHM of the shell areal density at stagnation.

TABLE I. Generalized formulas for the *degradation* of various implosion properties as a result of asymmetry. Here an assumption of a power-law reaction-rate is made, $\langle \sigma v \rangle \sim T^a$, where typically $3 < a < 4$. “Low-burn” refers to levels of yield amplification, Y_{amp} , less than 2 where the levels of alpha-heating nearly offset bremsstrahlung losses, making $PV^{5/3} \sim \text{const.}$ a good assumption for the hot-spot. Technically, the first three rows in the table below are for the “low-burn” limit. The fourth row below is a correction to be applied to the “low-burn” yield degradation for higher levels of yield amplification. Below χ_{1D} is the generalized Lawson parameter.¹⁷

Property	General asymmetry	Pure mode-1
P_{stag}	$\left(\frac{\rho \delta R_{WHM}}{\rho \delta R_{ave}} \right)^{5/2}$	$(1 - f^2)^{5/2}$
DSR, E_{hs}	$\frac{\rho \delta R_{WHM}}{\rho \delta R_{ave}}$	$(1 - f^2)$
$P\tau, T_{thermal}$	$\frac{\rho \delta R_{WHM}}{\rho \delta R_{ave}}$	$(1 - f^2)$
Yield,	$\left(\frac{\rho \delta R_{WHM}}{\rho \delta R_{ave}} \right)^a$	$(1 - f^2)^a$
$Y_{low-burn}$		
Y_{amp}	$\exp \left[-1.2(a-1) \chi_{1D}^{1.2} \left(1 - \frac{\rho \delta R_{WHM}}{\rho \delta R_{ave}} \right) \right]$	$\exp \left[-1.2(a-1) \chi_{1D}^{1.2} f^2 \right]$

Equation (14) explains why the formula for the mode-1 two-piston model appeared to accurately predict the yield degradation when applied to simulations that included mode-1, mode-2, and mode-4 as shown in Ref. 2 last year. Note that an alternate expression for RKE in the presence of asymmetry has been published by Springer *et al.*,⁹ which at lowest order in $\delta R/R_{ave}$ is equivalent to Eq. (14), but differs in higher order terms.

A. Application of the six-piston model for mode-2 and mode-2/1 coupling

In general, Eq. (5) or (7) can be applied to any arbitrary asymmetry to calculate Eq. (14) and the degradation factors in Table I. The utility of the six-piston model results of Sec. II can be demonstrated with a few case studies of a couple of specific asymmetry situations:

1. Axisymmetric mode-2 shell and mode-2 hot-spot

For this case of an axisymmetric mode-2 shell and mode-2 hot-spot asymmetry, the following parameterizations will be assumed in order to illustrate the physics of the coupling between the shell and hot-spot:

$$\rho \delta R = P_{0,\rho\delta R} \left[1 + \frac{1}{2} \frac{P_{2,\rho\delta R}}{P_{0,\rho\delta R}} (3 \cos^2 \theta - 1) \right], \quad (15)$$

$$R_{hs} = P_{0,hs} \left[1 + \frac{1}{2} \frac{P_{2,hs}}{P_{0,hs}} (3 \cos^2 \theta - 1) \right], \quad (16)$$

where the P 's in Eqs. (15) and (16) are the Legendre coefficients that fit the asymmetry of interest at the time of stagnation. For the six-piston model configuration of Fig. 2, the two parameterizations of Eqs. (15) and (16) are discrete ($\theta=0$, $\theta=\pi/2$, and $\theta=\pi$) and can be independent, but for application to an ICF implosion it should be recognized that total shell mass conservation links Eqs. (15) and (16).

By treating the discrete piston areas as $\sim R_{sh}^2$, we can assemble Eqs. (5) and (6) for the asymmetry described by Eqs. (15) and (16). Without a loss of generality, $P_{0,\rho\delta R}$ and $P_{0,hs}$ will be set to unity as a normalization in order to declutter the following math (these coefficients could be carried along only to find out that they will divide out in the final relationships). So, the six-pistons are characterized by

$$A_{z,-z} = \left(1 + \frac{P_{2,hs}}{P_{0,hs}} \right)^2, \quad (17)$$

$$\rho \delta R_{z,-z} = 1 + \frac{P_{2,\rho\delta R}}{P_{0,\rho\delta R}}, \quad (18)$$

$$A_{x,-x,y,-y} = \left(1 - \frac{1}{2} \frac{P_{2,hs}}{P_{0,hs}} \right)^2, \quad (19)$$

$$\rho \delta R_{x,-x,y,-y} = 1 - \frac{1}{2} \frac{P_{2,\rho\delta R}}{P_{0,\rho\delta R}}, \quad (20)$$

which allows construction of the key ratio of Eqs. (5) and (6),

$$\left. \frac{\rho \delta R_{WHM}}{\rho \delta R_{ave}} \right|_{mode-2} = \frac{\left[\left(\frac{P_{2,hs}}{P_{0,hs}} \right)^2 + 2 \right]^2 \left(2 - \frac{P_{2,\rho \delta R}}{P_{0,\rho \delta R}} \right) \left(1 + \frac{P_{2,\rho \delta R}}{P_{0,\rho \delta R}} \right)}{\left[4 + \left(\frac{P_{2,\rho \delta R}}{P_{0,\rho \delta R}} + 2 \right) \left(\frac{P_{2,hs}}{P_{0,hs}} \right)^2 + 4 \frac{P_{2,hs}}{P_{0,hs}} \frac{P_{2,\rho \delta R}}{P_{0,\rho \delta R}} \right] \left[\left(\frac{P_{2,hs}}{P_{0,hs}} \right)^2 - 2 \frac{P_{2,hs}}{P_{0,hs}} \frac{P_{2,\rho \delta R}}{P_{0,\rho \delta R}} + \frac{P_{2,\rho \delta R}}{P_{0,\rho \delta R}} + 2 \right]}, \quad (21)$$

which is perhaps more clear and useful when Taylor expanded in the limit of $P_2/P_0 \ll 1$ asymmetry (see Fig. 3),

$$\left. \frac{\rho \delta R_{WHM}}{\rho \delta R_{ave}} \right|_{mode-2} \approx 1 - \frac{1}{2} \left(\frac{P_{2,\rho \delta R}}{P_{0,\rho \delta R}} \right)^2 \left(1 + \frac{P_{2,hs}}{P_{0,hs}} \right) + \frac{1}{4} \left(\frac{P_{2,\rho \delta R}}{P_{0,\rho \delta R}} \right)^3 + \dots \quad (22)$$

This expansion emphasizes that the dominant contribution to RKE comes from the shell asymmetry, while the hot-spot asymmetry is a higher order correction. As seen in Fig. 3, for moderate levels of asymmetry the variation in the shell areal density dominates the contribution to RKE and yield degradation and in cases with no shell asymmetry, the hot-spot asymmetry has no impact.

However, for extreme asymmetry $P_{2,\rho \delta R}/P_{0,\rho \delta R} > 0.4$ Eqs. (21) and (22) break down (see Fig. 4). As seen in Fig. 4, using Eq. (5) directly does continue to be a good metric even for extreme levels of RKE , but when $RKE/(0.66KE) > 0.1$ the multivalued nature of Eq. (5) implies that Eq. (15) is no longer a good description of the asymmetry which is confirmed by noting that higher modes (concentrations of $\rho \delta R$ at the poles) become apparent in the simulations of these cases.

2. Axisymmetric mode-1 shell and mode-2 hot-spot

Here, we parameterize the mode-1 of the shell, in a fashion similar to Sec. II A 1. Namely,

$$\rho \delta R = P_{0,\rho \delta R} \left(1 + \frac{P_{1,\rho \delta R}}{P_{0,\rho \delta R}} \cos \theta \right), \quad (23)$$

while the hot-spot parameterization is the same as Eq. (16). The six-pistons are then characterized by

$$A_{z,-z} = \left(1 + \frac{P_{2,hs}}{P_{0,hs}} \right)^2, \quad (24)$$

$$\rho \delta R_{\pm z} = 1 \pm \frac{P_{1,\rho \delta R}}{P_{0,\rho \delta R}}, \quad (25)$$

$$A_{x,-x,y,-y} = \left(1 - \frac{1}{2} \frac{P_{2,hs}}{P_{0,hs}} \right)^2, \quad (26)$$

$$\rho \delta R_{x,-x,y,-y} = 1, \quad (27)$$

which allows construction of the key ratio of Eqs. (5) and (6). The full expression is unwieldy, so only the Taylor expanded version is given here

$$\left. \frac{\rho \delta R_{WHM}}{\rho \delta R_{ave}} \right|_{mode-1/2} \approx 1 - \left(\frac{1}{3} + \frac{2}{3} \frac{P_{2,hs}}{P_{0,hs}} \right) \left(\frac{P_{1,\rho \delta R}}{P_{0,\rho \delta R}} \right)^2 + \dots \quad (28)$$

Figure 5 plots the RKE and yield consequence of Eq. (28). As in the mode-2 shell asymmetry with a mode-2 hot-spot, the hot-spot asymmetry is a correction term. Interestingly, Eq. (28) implies that negative $P_{2,hs}/P_{0,hs}$ can be used to compensate for a z-aligned mode-1 in the shell.

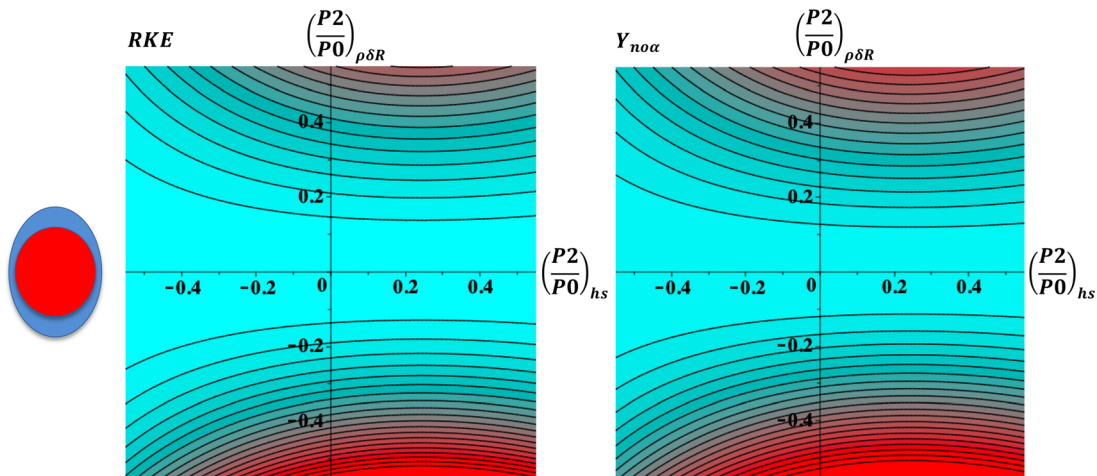


FIG. 3. The case of a mode-2 hot-spot and mode-2 shell as indicated by the figure on the left. Using Eq. (22), the expression for RKE , Eq. (14) (center) and no- α yield degradation from Table I (right) are plotted as functions of hot-spot and shell asymmetry. As can be seen in the figure, the majority of the variation is due to the shell asymmetry, albeit some variation due to the hot-spot asymmetry can be observed if the shell asymmetry is large enough. Contours in the left plot increase from 1% (cyan) to 20% (red) by 1% increments and decrease in the right plot from 97.5% (cyan) to 50% (red) by 2.5% increments.

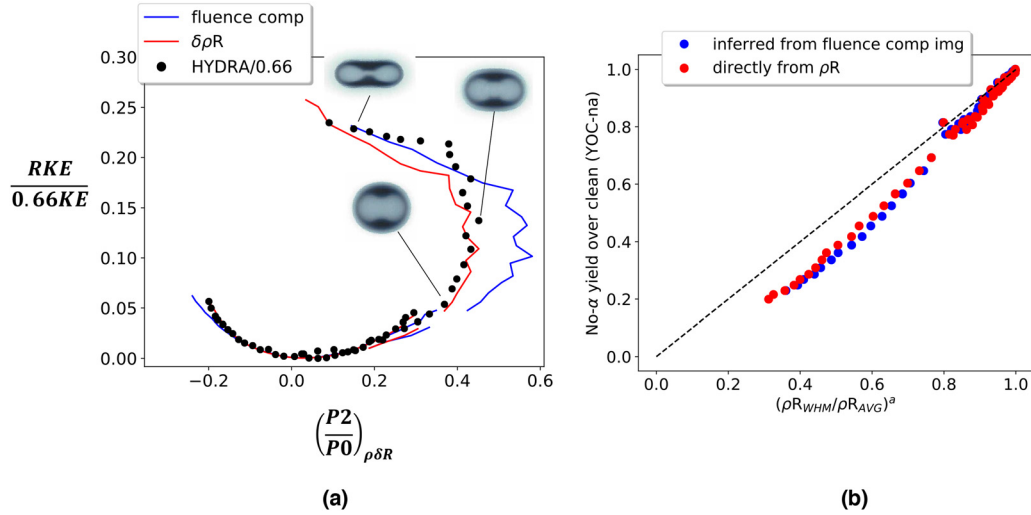


FIG. 4. (a) RKE calculations as a function of increasing radiation flux asymmetry are shown from 2D HYDRA simulations. The black points are the simulations calculation of RKE, while the red line is the WHM formula calculated using the simulated shell $\rho\delta R$ at stagnation, and the blue line the WHM formula applied to the synthetic fluence compensated neutron image from the simulation (a few examples are inset). (b) Using the same simulation data as in frame (a), the no-burn yield degradation is plotted vs the WHM metric with $a = 3.3$.

B. Mode-1 shell and mode-1 hot-spot, two-piston model cross-check and model limitations

If in Eq. (28) one takes $P_{2,hs}/P_{0,hs} = 0$, the two-piston model result, $RKE/(0.66KE) = (P_{1,\rho\delta R}/P_{0,\rho\delta R})^2$, should be recovered but instead $RKE/(0.66KE) = (1/3)(P_{1,\rho\delta R}/P_{0,\rho\delta R})^2$ is obtained. The same confounding result is obtained if the integral form, Eq. (7), is used instead of discrete pistons. The conundrum appears to be a consequence of having only two of the six pistons participating in the momentum and kinetic energy imbalance when a pure mode-1 perturbation is directed along one of the three primary axes of the

coordinate system,¹⁸ implying a subtle aspect of the area weighting needed in the piston model framework.

In the two-piston model, ρR and DSR were treated interchangeably, differing only by a constant conversion factor. The piston models are clearly dynamic models that depend upon the piston ρR . DSR is neutron scattering measurement, that by its nature samples a sector of the implosion shell. So, in calculating the WHM, Eq. (5), (6), or (7) to compare to simulations or data one should either use ρR with the area weighting factor included or instead calculate the harmonic mean (HM) and average with

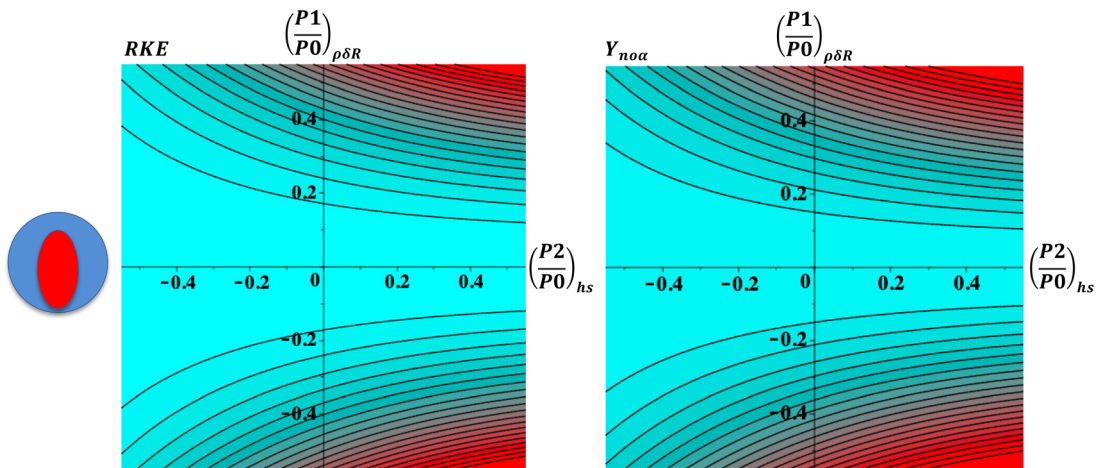


FIG. 5. The case of a mode-2 hot-spot and mode-1 shell (or equivalently and mode-2+mode-1 hot-spot) as indicated by the figure on the left. Using the WHM formula, the expression for RKE (center) and no- α yield degradation (right) are plotted as functions of hot-spot and shell asymmetry. As can be seen in the figure, the majority of the variation is due to the shell asymmetry, albeit some variation due to the hot-spot asymmetry can be observed if the shell asymmetry is large enough. Contours in the left plot increase from 1% (cyan) to 20% (red) by 1% increments and decrease in the right plot from 97.5% (cyan) to 50% (red) by 2.5% increments. Note these plots are obtained from the full expression that lead to the Taylor expansion shown in Eq. (28).

no area weighting when using *DSR* (since *DSR* is already effectively weighted by area).

1. Application of WHM metric to NIF data

In cases where mode-1 asymmetry is dominant, it has already been understood that hot-spot drift velocity, v_{hs} , inferred from neutron time-of-flight (NToF) diagnostics is a reasonably good signature for the asymmetry^{1,19} and that by forming the ratio $f = v_{hs}/v_{imp}$ implosion performance degradation can be understood (e.g., the right column of Table I). Since even-modes of asymmetry result in no net center-of-mass motion, v_{hs} is not a good metric for those modes, but mode-2 can have an impact on v_{hs} .²⁰ Neutron diagnostics, other than NToF, turn out to be useful for assessment of general implosion asymmetry beyond mode-1.

Radiochemical nuclear activation detectors (NADs) arrayed around the NIF target chamber record the time-integrated local neutron fluence during implosion experiments. The original form of this set of diagnostics was called the flange nuclear activation detectors (FNADs)²¹ which was later replaced by the real-time nuclear activation detectors (RTNADs).²² The principle of the diagnostic is that neutrons generated by the DT fusion in the hot-spot of an ICF implosion must scatter off the compressed fuel surrounding the hot-spot on their way to the NADs. The probability of neutron scattering is sensitive to the areal density that a fusion neutron encounters as it leaves the hot-spot, so a low-mode map of the 4π areal density of the shell surrounding the hot-spot can be reconstructed from NAD data using a neutron transport calculation. For a given activation detector, subtending a small solid angle, a high level of activation would imply a relatively low areal density along the radius connecting the implosion center to the detector, while a low level of activation would imply a relatively high areal density along the same radius.

Thus, with the FNAD or RTNAD data along with a neutron transport calculation, Eqs. (7) can be computed and the expressions of Table I can be used to calculate various implosion performance degradation factors. While it is tempting to drop the area weighting in Eqs. (7) and just integrate over the solid angle, this simplification yields a poor approximation generally, so information about the hot-spot shape, from hot-spot imaging diagnostics,²³ and fluence compensated neutron imaging²⁴ is needed in order to capture the important effect of the area weighting (a paper detailing the integration of the various experimental diagnostics will be forthcoming).

The data inferred impact, using the procedure described above, of low-mode asymmetry upon the alpha-heated yield of DT implosions on NIF is shown in Fig. 6. The data used here is derived from the fluence compensated neutron image from a single line-of-sight (90–315, where 90 refers to the polar angle and 315 the azimuthal angle coordinates on the NIF facility), including area weighting. This method works best for 2D asymmetries, as 3D asymmetries require more line-of-sight to properly reconstruct or the inclusion of other data (e.g., RTNAD's). Additional neutron image lines-of-sight are being deployed, and methods to use the RTNAD data are also under development. The degradations shown also include the effects of alpha heating (expression for Y_{amp} in Table I). As can be observed in Fig. 6, most DT shots on NIF have been negatively impacted by asymmetry, some as noted by their shot numbers, very severely. The typical alpha-heated yield degradation is calculated to be 0.66 ± 0.14 (1σ RMS) of

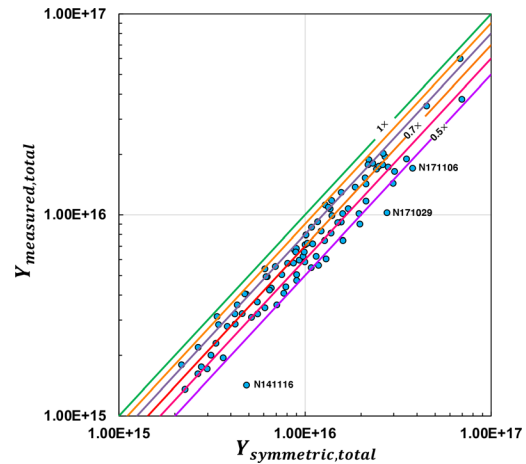


FIG. 6. For indirect-drive DT implosion experiments on NIF up to February 2021 with total neutron yield $>10^{15}$ and using fluence compensated neutron data²⁴ to calculate Eq. (7), the measured neutron yield is plotted against the yield for the equivalent symmetric implosion, as inferred from the 6-piston model (the $Y_{low-burn}$ and Y_{amp} expressions in Table I). The contours show by how much the fusion yield was degraded by asymmetry. From the plot, it's clear that most implosions on NIF have not been symmetric and that the level of asymmetry that was present in most shots had a measurable impact upon fusion yield. A few shots with yields degraded by asymmetry by more than a factor of $2\times$ are noted for reference.

what an identical, but symmetric, implosion would have been. While retrospective, Fig. 6 and the underlying data imply that asymmetry has been a significant factor in the performance of NIF DT implosions for many years, justifying the steps that have been taken to understand the sources, impact, and mitigation of asymmetry.

III. COAST-TIME, HOHLRAUM COOLING, AND THE CONNECTION TO PISTON-MODEL STAGNATION PRESSURE

This section, along with Appendix C, focuses more upon the one-dimension (1D) physics of the piston model. In particular, it appears from Eq. (10) that regardless of asymmetry considerations the hot-spot pressure at peak velocity, P_{pv} , and volume, V_{pv} (or equivalently radius at peak velocity R_{pv}) are parameters that can be used to modify the stagnation pressure and other hot-spot properties. Here, the connection between the implosion properties at peak velocity and coast-time¹⁴ is made in order to show how reduced coast-time can be used to increase stagnation pressure and other desirable hot-spot properties.

Consistent with the piston model abstraction, and to simplify the mathematics in favor of understanding, here the shell of the implosion is effectively treated as incompressible and thin (so that the hot-spot radius and radius of the shell COM are not very different). Mention of where these assumptions matter will be pointed out when relevant to key results. As will be seen, the essential physical picture is that R_{pv} is defined by the radius at which the declining (due to cooling) late-time ablation pressure outside the implosion balances the increasing (due to convergence) hot-spot pressure inside the implosion.

As was previously established, reducing coast-time (the time between peak ablation pressure and implosion bang-time) generally results in a significant increase in hot-spot performance parameters

such as stagnation pressure, hot-spot temperature, yield, etc.^{14,25–28} The benefit of reduced coast-time on average compression as measured by *DSR* has also been established for low-foot implosions,^{29–32} but hydro-instability and mixing frustrated proper hot-spot performance for the low-foot implosions.

Linking coast-time to hohlraum cooling using a simple bi-linear rise and fall of ablation pressure (see Fig. 7) enabled an analytic solution of a rocket model for implosion acceleration in terms of three dimensionless parameters. One of the dimensionless parameters was the ratio of the rate of ablation pressure increase as the hohlraum heats to peak radiation temperature, T_{rad} , over the rate of ablation pressure decrease as the hohlraum cools. In Ref. 14, the physical picture was that the late-time ablation pressure on a short coast-time implosion can be significantly higher than that same implosion with a longer coast-time during the cooling phase of the hohlraum (this is reflected in the left frame of Fig. 7). As a result of the relatively larger ablation pressure with a shorter coast-time, the implosion acquires a somewhat larger velocity and decompression of the outer surface of the shell is delayed, resulting in effectively more shell ρv_{imp}^2 and a more compressed state at stagnation.

The useful parameterization of ablation pressure was¹⁴

$$p_{abl}(t) = \begin{cases} p_{abl,max} \frac{t}{t_{peak}}, & t_{rise} \leq t \leq t_{peak}, \\ p_{abl,max} \frac{t_{peak} + \tau_{cool} - t}{\tau_{cool}}, & t \geq t_{peak}, \end{cases} \quad (29)$$

where τ_{cool} is the “cooling time.” Note that in Eq. (29) the time t_{peak} is the time of peak ablation pressure, not the time of peak velocity. Parameterizations other than the linear cooling of Eq. (29) could also be assumed (e.g., exponential), with little impact on the essential physical picture. Neglecting the sound wave transit time across a thick compressible shell, the equation of motion for the deceleration of shell of

the implosion is determined by the ablation pressure pushing the shell inwards and the internal hot-spot pressure pushing the shell outwards, namely,

$$\frac{d\dot{R}}{dt} \approx \frac{P(R) - p_{abl}(t)}{\rho \delta R_{ave}}, \quad (30)$$

where R is the 1D shell radius.

In order to connect the radius and pressure at peak velocity of the piston model (which is a deceleration model that neglects the ablation pressure after the time of peak velocity) to the ablation pressure history and, in particular the coast-time, the full solution of Eq. (30) is not required. Instead, it is simply noted that peak velocity must occur when the shell acceleration is zero ($\ddot{R} = 0$) and the pressures inside and outside the shell balance

$$P(R_{pv}) = P_{pv} = p_{abl}(t_{pv}) \approx p_{abl,max} \frac{t_{peak} + \tau_{cool} - t_{pv}}{\tau_{cool}}, \quad (31)$$

where t_{pv} is the time of peak velocity and here it has been assumed that $t_{pv} > t_{peak}$. Consistent with the adiabatic hot-spot assumption of the piston model, $P(R) = P_{pv}(R_{pv}/R)^5$, it's clear that the left hand side of Eq. (31) is P_{pv} .

The deceleration time from the time of peak velocity to the time of minimum volume, $t_{decel} = t_{minV} - t_{pv}$, is given by the solution of Eq. (8) and is nearly R_{pv}/v_{imp} (explicit solution of Eq. (8) for time gives this with a complicated pre-factor that is of order unity to within 12%) as shown in Ref. 2, thus

$$t_{pv} \approx -\frac{R_{pv}}{v_{imp}} + t_{minV}. \quad (32)$$

The stagnation pressure from the piston model is related to the pressure and radius at peak velocity, namely, using an alternate expression for the piston model pressure, yet is equivalent to Eq. (10),

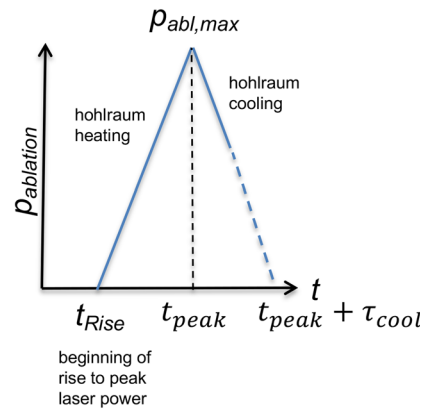
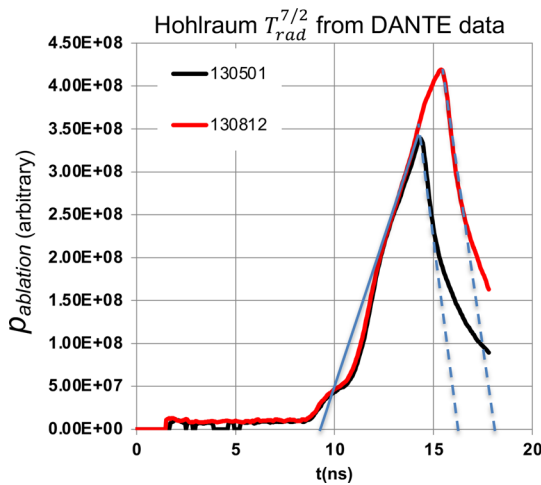


FIG. 7. (Left) Shown are plots of $T_{rad}^{7/2}$, as measured by a low energy resolution spectrometer on an arbitrary scale, for a representative long coast-time implosion experiment (black) and a shorter coast-time implosion (red). As can be seen and as has been noted before, the shape of the profiles is roughly triangular. The rise and fall of T_{rad} are a reflection of the prescribed laser power time dependence and hohlraum energy losses. (Right) The idealized rise and fall of ablation pressure, which scales as $T_{rad}^{7/2}$ for x-ray driven implosions with constant ablator albedo, is assumed bi-linear for ease of analysis. The key times of interest are noted. Here, the early-time decline of ablation pressure is modeled as linear, but exponential decay or other functions could also be assumed to better capture the late-time cooling implied by the data shown in the left frame. (Adapted from Ref. 14).

$$P_{stag} = \frac{\rho \delta R_{WHM} v_{imp}^2}{R_{hs}} = P_{pv} \left(\frac{R_{pv}}{R_{hs}} \right)^5, \quad (33)$$

where the second relation in the above equation is the adiabatic assumption which has been used repeatedly in the context of the piston model. Finally, the coast-time, t_{coast} , is defined as the time difference between stagnation and peak ablation pressure

$$t_{coast} = t_{minV} - t_{peak}. \quad (34)$$

Equations (31)–(34) form a set of equations that can be used to eliminate P_{pv} , t_{pv} , and t_{minV} in favor a relationship between R_{pv} and t_{coast} . One finds

$$R_{pv}^6 - v_{imp}(t_{coast} - \tau_{cool})R_{pv}^5 - R_{nc}^6 = 0, \quad (35)$$

where R_{nc} is defined as the “no-coast radius”

$$R_{nc} = \left(\frac{\rho \delta R_{WHM} v_{imp}^3 \tau_{cool} R_{hs}^4}{P_{abl,max}} \right)^{1/6}. \quad (36)$$

It is interesting to note that $R_{nc} = (P_{stag}/P_{abl,max})^{1/6} \times [(v_{imp} \tau_{cool}) R_{hs}^5]^{1/6}$, which is geometric-like average between the length-scale $v_{imp} \tau_{cool}$ and R_{hs} amplified by sixth root of the ratio of stagnation pressure to peak ablation pressure. If shell mass is conserved between peak velocity and stagnation we can relate R_{hs} to $1/v_{imp}$ using Eq. (C2) and see that R_{nc} is very nearly a constant (i.e., has only a very weak dependence upon v_{imp}) for a fixed design. Namely,

$$R_{nc} = \left(\frac{c_1 v_{imp} \tau_{cool}}{P_{abl,max}} \right)^{1/6}, \quad (37)$$

where c_1 is a constant related to the hot-spot entropy at peak velocity.

While a general closed-form solution of Eq. (35) is not possible, the equation clearly relates t_{coast} to R_{pv} and other implosion properties. Noting that for most practical designs on the NIF, that t_{coast} and τ_{cool} comparable,³⁴ a useful approximate solution to Eq. (35) can be obtained.

In the limit that $(t_{coast} - \tau_{cool})v_{imp}/R_{nc}$ is a small parameter of order $\mathcal{O}(\epsilon)$, Eq. (35) can be solved by Taylor series expansion of $R_{pv} \approx R_{pv}^{(0)} + R_{pv}^{(1)} + \dots$, where $R_{pv}^{(1)}/R_{pv}^{(0)} \sim \mathcal{O}(\epsilon)$ and equating terms order-by-order. One obtains

$$R_{pv} \approx R_{nc} + \frac{1}{6} v_{imp}(t_{coast} - \tau_{cool}), \quad (38)$$

where it is evident that R_{pv} decreases linearly with decreasing t_{coast} . This analytic approximation appears to well fit numerical solutions (see Fig. 8) from a combined rocket plus adiabatic hot-spot model.³³

Two other approximate solution limits of Eq. (35) are possible. When $(t_{coast} - \tau_{cool})v_{imp}/R_{nc} \gg 1$, then the solution of Eq. (35) is dominated by the two left terms in the equation, so $R_{pv} \approx v_{imp}(t_{coast} - \tau_{cool})$ in this limit. When $(t_{coast} - \tau_{cool})v_{imp}/R_{nc} \ll -1$, then the solution of Eq. (35) is dominated by the right two terms in the equation, so $R_{pv} \approx R_{nc}^6/[v_{imp}(\tau_{cool} - t_{coast})]^{1/5}$. Figure 9 summarizes with a plot of Eq. (35) and the above approximations to the solution. It is clear from Fig. 9 that if $t_{coast} > \tau_{cool}$ reducing t_{coast} rapidly reduces R_{pv} , but if $t_{coast} < \tau_{cool}$, R_{pv} becomes much less responsive to changes in t_{coast} .

The implication of Eq. (35) and Fig. 9 is that the key benefit of reducing t_{coast} is that R_{pv} is reduced, which in turn increases the

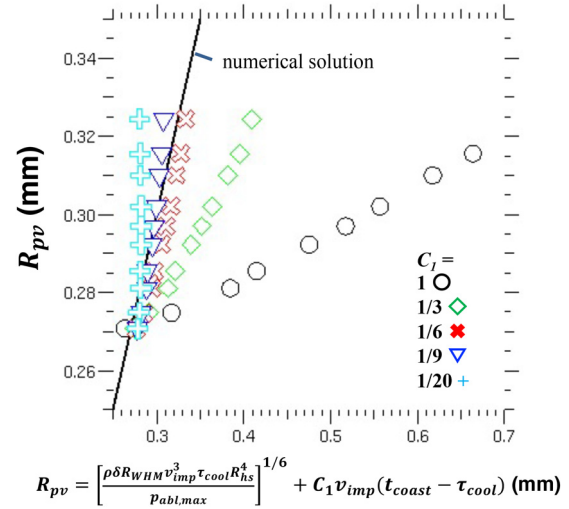


FIG. 8. A plot of R_{pv} calculated from equations of the form of Eq. (38), with the coefficient on the second term left variable, are compared to numerical solution from the scoping model of Ref. 33. In the numerical model, the laser energy varies from 1.3–1.85 MJ at 470 TW, the ablator thickness is 77 μm , the DT fuel layer is 55 μm , and a 6.72 mm diameter hohlraum is assumed with a fixed $\tau_{cool} = 1.88$ ns. As can be seen in the plot, Eq. (38), is an adequate approximation.

stagnation pressure (see Fig. 10), by decreasing t_{decel} and increasing the rate of converting kinetic energy into internal energy. As a result, the effective gravity, $g_{eff} = v_{imp}/t_{decel} = v_{imp}^2/R_{pv}$, that the shell experiences as it decelerates is likewise increased, but over a shorter time, which may have consequences for hydro-instability.³⁵

IV. CONCLUSION

The piston model presented in this paper is a very abstracted picture of a real implosion, yet it provides significant insight into what aspects of asymmetry are critical and it compares favorably to

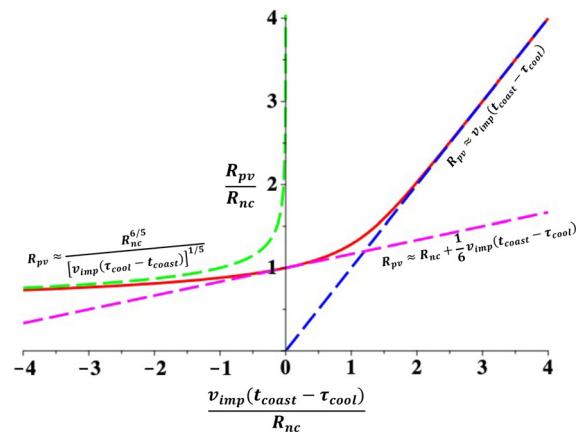


FIG. 9. A plot of R_{pv} calculated from Eq. (35) (solid red curve) is shown along with the approximations discussed in the text that are useful in various limits (dashed curves). In dimensionless form, Eq. (35) is a single universal curve. The parameter R_{nc} is defined in Eq. (36).

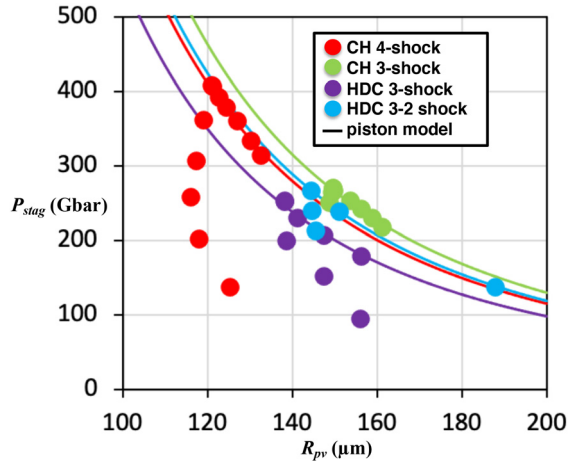


FIG. 10. 1D HYDRA^{36,37} simulations of various hypothetical NIF implosions varying coast-time only show the expected $1/R_{pv}^{5/2}$ behavior of stagnation pressures as predicted by Eq. (C6), while for implosions that do not maintain adiabat and final shock strength due to varying peak power and hence T_{rad} have points that falloff the curves.

simulations and provides a basis for understanding data. The key metric of asymmetry pointed out by the model is the ratio of shell areal density weighted harmonic mean average, Eq. (5), to average areal density, Eq. (6).

The degradation of all hot-spot properties and increase in RKE are simple functions of $\rho\delta R_{WHM}/\rho\delta R_{AVE}$. This model can be used to understand single modes of asymmetry or non-linear mode coupling. The piston model can be coupled with neutron data, that reconstruct the asymmetry of the shell, to explore the impact of asymmetry on actual implosions.

Extension of the two-piston model to include time-dependent asymmetry (see Appendix D) implies that hot spot performance is dominated by the shell asymmetry of the implosion at stagnation, rationalizing why the time-independent piston model seems to have some predictive capability even when compared to full simulations of implosions that have time-dependent asymmetry evolution. Time-dependent asymmetries are apparently perturbative corrections to the piston model solution, except in situations where the stagnation pressures are low.

Since the piston model framework considers the implosion from peak velocity to stagnation, it originally appeared that the radius at peak velocity was an independent parameter. However, upon further consideration it was clear that the radius of peak velocity is directly related to concept of coast-time and that R_{pv} is the radius at which the dropping (due to hohlraum cooling) ablation pressure outside the shell of the implosion matches the increasing (due to convergence) hot-spot pressure inside the implosion. Coupling the piston model picture to concept of coast-time reveals that R_{pv} is a rather important parameter that has significant leverage on stagnation pressure and other hot-spot performance quantities. The theory, Eq. (35), and supporting simulations imply that ICF implosion performance can be significantly improved if R_{pv} is minimized, but the benefit of the effect drops after $R_{pv} < R_{nc}$ which is

coincident with the coast-time dropping below the characteristic timescale of ablation pressure drop.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX A: PROOF THAT $\rho\delta R_{WHM} \leq \rho\delta R_{ave}$

The Cauchy-Schwarz inequality in an n -dimensional vector space, states that for any pair of real-space ($\Re$) vectors u_i and v_i the following inequality must always hold:

$$\left(\sum_{i=1}^n u_i v_i\right)^2 \leq \left(\sum_{i=1}^n u_i^2\right) \left(\sum_{i=1}^n v_i^2\right). \quad (\text{A1})$$

By inspection, if one equates $u_i = \sqrt{(\rho\delta R)_i} A_i$ and $v_i = \sqrt{A_i/(\rho\delta R)_i}$ then Eq. (A1) becomes, with $n=6$ for the six-piston model implosion,

$$\left(\sum_{i=1}^6 A_i\right)^2 \leq \left(\sum_{i=1}^6 (\rho\delta R)_i A_i\right) \left(\sum_{i=1}^6 \frac{A_i}{(\rho\delta R)_i}\right). \quad (\text{A2})$$

One can now identify the weighted harmonic-mean areal density average and areal density average, by dividing both sides of Eq. (A2) by $\sum_{i=1}^6 A_i/(\rho\delta R)_i$ and by $\sum_{i=1}^6 (\rho\delta R)_i A_i$. It can then be seen that $\rho\delta R_{WHM} \leq \rho\delta R_{ave}$ regardless of the arrangement of shell areal density asymmetry.

APPENDIX B: HOT-SPOT VELOCITY AND SHELL CENTER-OF-MASS VELOCITY

Assuming all six-pistons of two are initially moving toward each other with implosion speed v_{imp} , the center of mass velocity of the system is given by

$$\vec{v}_{hs} = \dot{X}_{CM}\hat{x} + \dot{Y}_{CM}\hat{y} + \dot{Z}_{CM}\hat{z}. \quad (B1)$$

The component of each unit vector direction are given by

$$\dot{X}_{CM} = \frac{1}{m_{shell}}(\rho\delta R_{-x} - \rho\delta R_{+x})A_x v_{imp} \quad (B2)$$

$$\dot{Y}_{CM} = \frac{1}{m_{shell}}(\rho\delta R_{-y} - \rho\delta R_{+y})A_y v_{imp} \quad (B3)$$

$$\dot{Z}_{CM} = \frac{1}{m_{shell}}(\rho\delta R_{-z} - \rho\delta R_{+z})A_z v_{imp} \quad (B4)$$

with the total mass being $m_{shell} = \sum_j \rho\delta R_j A_j$ with the summation index running over the six pistons. As in the two-piston model, the assertion is that the center-of-mass velocity, Eq. (B1), is equivalent to the hot-spot center of geometry velocity.

APPENDIX C: SOME ADDITIONAL 1D PISTON MODEL RELATIONSHIPS

In this section, we note a handful of useful 1D implosion relationships that follow from a few simple physics considerations and note the equivalence to some of the piston model relationships. The relationships are useful for the coast-time discussion in Sec. IV and for a better understanding of the 1D scalings suggested by the piston model.

Starting with conservation of energy from peak velocity to stagnation, $m_{shell}v_{imp}^2/2 = 3P_{stag}V_{min}/2$ (neglecting the internal energy of the shell at stagnation), with hot-spot volume $V_{min} = 4\pi R_{hs}^3/3$, an expression for stagnation pressure is obtained

$$P_{stag} = \frac{1}{4\pi} \frac{m_{shell}v_{imp}^2}{R_{hs}^3}. \quad (C1)$$

which we can see is equivalent to the 1D version of Eq. (11) [which is also the left equality in Eq. (33)] by identifying the shell mass as $m_{shell} \approx 4\pi(\rho\delta R)_{stag}R_{hs}^2$, where $(\rho\delta R)_{stag}$ is the shell areal density at stagnation.

Again using the adiabatic assumption $P_{stag}R_{hs}^5 = P_{pv}R_{pv}^5 = c_1$ (c_1 being a constant) with Eq. (C1) above yields

$$c_1 = P_{pv}R_{pv}^5 = \frac{1}{4\pi} m_{shell}v_{imp}^2 R_{hs}^2. \quad (C2)$$

Assuming little differential ablation of the shell occurs as when coast time varies, as shown by 1D simulations, means m_{shell} at peak velocity can be approximated as constant for a fixed design. On the other hand, v_{imp} does vary some with coast time¹⁴ as shown in Fig. 11, a consequence of the Rocket model for which $v_{imp} \sim v_{exhaust} \sim \sqrt{T_{rad}}$, even for fixed mass remaining. So Eq. (C2) implies that the product $v_{imp}R_{hs} \approx \sqrt{(4\pi c_1/m_{shell})}$ is also constant for a fixed design, a conclusion that is supported by simulations (see Fig. 11). Remarkably, the interchangeability of v_{imp} and $1/R_{hs}$ is just a consequence of the adiabatic assumption and energy conservation.

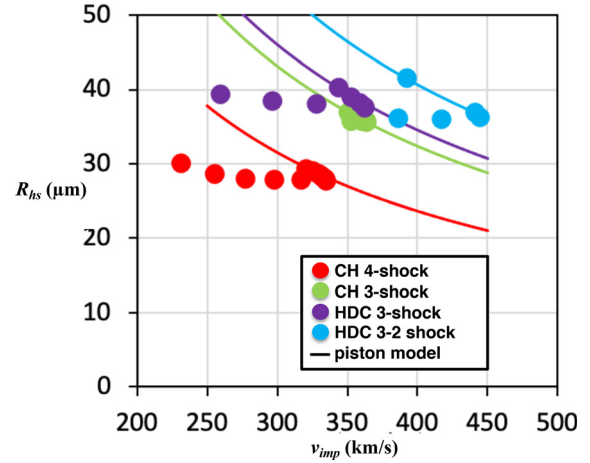


FIG. 11. 1D HYDRA^{36,37} simulations of various hypothetical NIF implosions varying coast-time only show the expected scaling of R_{hs} vs v_{imp} from Eq. (C2) while implosions that do not maintain adiabat and final shock strength due to varying peak power and hence peak T_{rad} have points that falloff the curves.

With $v_{imp}R_{hs}$ constant, the stagnation pressure, Eq. (C1), simplifies to

$$P_{stag} = \left(\frac{m_{shell}}{4\pi}\right)^{5/2} \frac{v_{imp}^5}{c_1^{3/2}} \sim v_{imp}^5. \quad (C3)$$

Equation (C3) is also supported by simulations (see Fig. 12) and has been noted in DT implosion data on numerous occasions. An important consequence of Eq. (C3) is that the $\rho\delta R$ in Eq. (10), which shows the same v_{imp}^5 scaling must be interpreted as the $\rho\delta R$ at peak velocity, not stagnation, in order to be consistent with mass conservation ($\rho\delta R R^2 \sim \text{const}$) when compared to a convergent simulations or data [this is *not* true for Eq. (11) or (33), because of

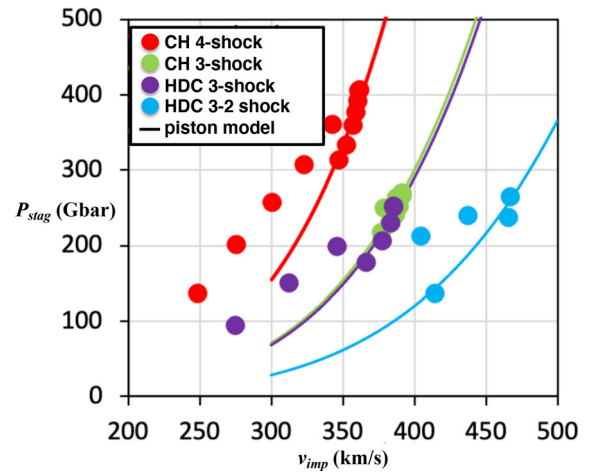


FIG. 12. 1D HYDRA^{36,37} simulations of various hypothetical NIF implosions varying coast-time only show the expected scaling of P_{stag} vs v_{imp} from Eq. (C3) while implosions that do not maintain adiabat and final shock strength due to varying peak power and T_{rad} have points that falloff the curves.

Eq. (C1)]. That is, re-writing the middle expression in Eq. (C3) using $m_{shell} \approx 4\pi(\rho\delta R)_{pv}R_{pv}^2$, where $(\rho\delta R)_{pv}$ is the shell areal density at peak velocity gives

$$P_{stag} = \left[\frac{(\rho\delta R)_{pv}}{R_{pv}} \right]^{5/2} \frac{v_{imp}^5}{P_{pv}^{3/2}}, \quad (C4)$$

which looks like the symmetric implosion version of Eq. (10) if one notes the peak velocity vs stagnation distinction in the areal density, a distinction that is outside the scope of the piston model abstraction.

Another physical process beyond the scope of the piston model abstraction, but present in real/simulated ICF implosions are shocks and the passage of a strong converging shock in the hot-spot gas sets P_{pv} and has a subtle impact upon the connection of R_{pv} to other implosion variables. The appearance of $R_{pv}^{5/2}$ in the denominator of Eq. (C4) begs the question of how variation of R_{pv} impacts stagnation pressure?

Asserting that the timescale for a shock wave to travel from R_{pv} to $R=0$, R_{pv}/c_s (c_s being the sound speed in the shocked hot-spot) needs to be proportional to the dynamic timescale R_{hs}/v_{imp} gives an interesting relationship,

$$R_{pv} \sim c_s R_{hs} / v_{imp} \sim \frac{1}{v_{imp}^2} \sim R_{hs}^2, \quad (C5)$$

where Eq. (C2) has been used to give the two scalings on the right-hand-side of Eq. (C5). Another way to interpret C5 is that the adiabatic assumption, $PR^5 \sim \text{const.}$, cannot be valid until the shock processes the hot-spot plasma. Conversely, if additional shocks enter the hot-spot plasma after peak velocity, $PR^5 \sim \text{const.}$ would not apply (rather obviously) and the piston model relationships might not be valid.

Using Eq. (C5) in Eq. (C3) then means

$$P_{stag} \sim \left(\frac{m_{shell}}{4\pi} \right)^{5/4} \frac{c_s^{5/2}}{c_1^{1/4}} \frac{1}{R_{pv}^{5/2}}. \quad (C6)$$

So for “real” implosions we are lead to the conclusion that $R_{hs} \sim \sqrt{R_{pv}} \sim 1/v_{imp}$ for a fixed design (e.g., adiabat, target geometry, etc.), which indeed appears to be the usual case according to simulations (see Fig. 13). The key result being that P_{stag} scales as $\sim v_{imp}^5$, as shown in Eq. (C3), or as $\sim 1/R_{hs}^5$ or as $\sim 1/R_{pv}^{5/2}$. Because of shell mass conservation between peak velocity and stagnation in convergent geometries and conservation of hot-spot entropy post-shock, the parameters v_{imp} , R_{hs} , and R_{pv} are not independent as they appear in the piston model equations, but are in fact linked by proportionality constants for a fixed implosion design.

On a final comment, examining the quantity $P_{stag}^2 E_{stag}$, which is a measure of progress toward ignition one finds using the above (no-burn) relations that

$$P_{stag}^2 E_{stag} = 2\pi(P_{stag}R_{hs})^3 \sim \left(\frac{m_{shell}c_s^2}{4\pi} \right)^3 \frac{2\pi}{R_{pv}^6}, \quad (C7)$$

which is a very strong scaling, indicating that the radius of peak velocity has tremendous leverage on progressing toward ignition. Furthermore the similarity between $m_{shell}c_s^2/R_{pv}^2$ above and the intuited capsule design metric Eq. (16) of Ref. 33, but for a swap of c_s for v_{imp} , is notable.

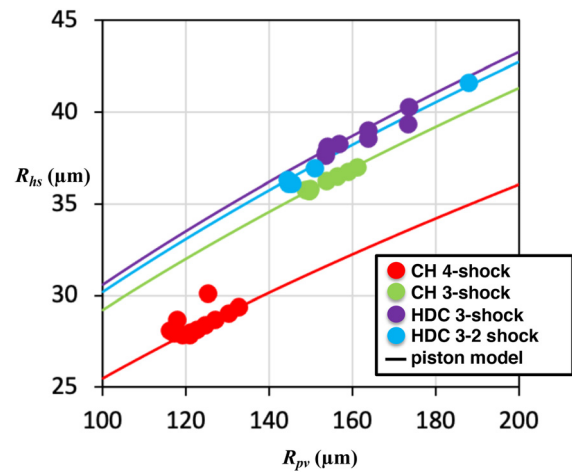


FIG. 13. 1D HYDRA^{36,37} simulations of various hypothetical NIF implosions varying coast-time only show the expected scaling of R_{hs} vs R_{pv} from Eq. (C5) while implosions that do not maintain adiabat and final shock strength due to varying peak power and hence T_{rad} have points that falloff the curves.

APPENDIX D: TWO-PISTON MODEL FOR TIME-DEPENDENT MODE-1 ASYMMETRY SWING

The original piston model² treated the piston asymmetry (as shown in Fig. 1) in $\rho\delta R$ as constant in time, so the shell asymmetry fraction mentioned in the introduction, f , was assumed constant. The fact that the original two-piston model worked well when compared to data and simulation implies that the asymmetry of the configuration at the time of stagnation was the dominant degradation to an implosion. In this section, the consequence of adding a time dependence to mode-1 asymmetry (termed “swing”) is explored, not for the purpose of providing an accurate time-dependent asymmetry model, but instead to rationalize why the time-independent piston model works well even when compared to data and simulations that have time-dependent asymmetry.

The analog of Eq. (4) for a pure mode-1 asymmetry published in Ref. 2 and pictured in Fig. 1 is

$$\ddot{S} = \frac{P_{pv}}{\rho\delta R_{ave}} \left(\frac{2}{1-f^2} \right) \left(\frac{S_{pv}}{S} \right)^{5/3}. \quad (D1)$$

Like Eq. (4), Eq. (D1) is autonomous if $f = f(S)$. So, in order to recover an analytical solution to Eq. (D1) while also including asymmetry swing the following parameterization suggests itself:

$$f = f_{stag} + f'[S(t) - S_{min}], \quad (D2)$$

where f_{stag} is the same as Eq. (1) and is the asymmetry fraction at stagnation when the hot-spot diameter, $S = S_{min}$, is minimum and $f' = (f_{pv} - f_{stag})/(S_{pv} - S_{min})$ is a measure of swing. Equation (D2) is a Taylor expansion that approximates the asymmetry fraction evolution, going from a value of $f_{stag} + f'(S_{pv} - S_{min})$ to a value of f_{stag} linearly with S . Higher order terms could, of course, be included in Eq. (D2), but at the cost of greatly increased complexity in the

solution. Note that a physical requirement that the two model piston masses be nonzero, implies a constrain on Eq. (D2), namely, $f_{stag}^2 + [f'(S_{pv} - S_{min})]^2 < 1$.

The solution of Eq. (D1) with Eq. (D2) is accomplished by noting $\dot{S} = \dot{S}(dS/dS)$ and series expanding $2/(1-f^2) \approx 2(1+f^2)$ (without which the forthcoming quadrature is not analytic). The quadrature solution then follows the same steps as Eqs. (8) and (9) above yielding, after a little algebra,

$$\left(\frac{S_{pv}}{S_{min}}\right)^{2/3} \left[1 - 6 \frac{f' S_{pv} f_{stag} S_{min}}{1 + f_{stag}^2 S_{pv}} + \frac{9 (f' S_{pv})^2}{2(1 + f_{stag}^2)} \left(\frac{S_{min}}{S_{pv}}\right)^2 \right] \approx \frac{2\rho\delta R_{ave} v_{imp}^2}{3P_{pv} S_{pv} (1 + f_{stag}^2)} + \dots \quad (D3)$$

where in obtaining Eq. (D3) the ICF limit of $\rho v_{imp}^2 \gg P_{pv}$ was used. For the sake of compactness in the following equations, it is convenient to define the dimensionless parameter $b = 3P_{pv} S_{pv} / (2\rho\delta R_{ave} v_{imp}^2) \ll 1$ noting that b is the ratio of hot-spot energy at peak velocity and shell kinetic energy at peak velocity.

Using the adiabatic assumption for pressure, $P_{stag}/P_{pv} = (S_{pv}/S_{min})^{5/3}$ appropriate for the two-piston geometry of Fig. 1, allows Eq. (D3) to be rewritten as a non-linear implicit equation for stagnation pressure. Namely,

$$\left(\frac{P_{stag}}{P_{pv}}\right)^{2/5} \approx \frac{1}{b(1 + f_{stag}^2)} + 6 \frac{f' S_{pv} f_{stag}}{1 + f_{stag}^2} \left(\frac{P_{pv}}{P_{stag}}\right)^{1/5} - \frac{9 (f' S_{pv})^2}{2(1 + f_{stag}^2)} \left(\frac{P_{pv}}{P_{stag}}\right)^{4/5}. \quad (D4)$$

Note that when $f' = 0$ in Eq. (D4), the expression for the piston pressure of Ref. 2 is recovered. Equation D4 is an implicit equation for P_{stag}/P_{pv} in terms of three dimensionless parameters: the

one-dimensional physics parameter (b), the measure of asymmetry at stagnation (f_{stag}), and a measure of asymmetry swing ($f' S_{pv}$). A plot of Eq. (D4) is shown in Fig. 14 for two different values of b .

By looking at Eq. (D4) one can see that the swing terms become small contributions to the equation in the limit of $P_{stag} \gg P_{pv}$, which is also confirmed by the right frame of the plot shown in Fig. 14 where $P_{stag}/P_{pv} \sim 100$'s. For cases where the increase in stagnation pressure is less extreme, $P_{stag}/P_{pv} \sim 10$'s, swing can have an impact as shown in the left frame of Fig. 14. The impact of swing as predicted by Eq. (D4) is consistent with intuition, namely, swings that make the implosion more symmetric by stagnation are helpful, while swings that make the implosion less symmetric at stagnation are more damaging. In practice, ICF implosions on NIF typically occupy a parameter space in between the two cases shown in Fig. 14.

Since $P_{stag}/P_{pv} \gg 1$ is the parameter range of interest, Eq. (D4) can be used to generate an iterative series solution for P_{stag} that is accurate and eliminates the implicit nature of Eq. (D4). The first few terms of the series solution to Eq. (D4) are

$$\frac{P_{stag}}{P_{pv}} \approx \left[\frac{1}{b(1 + f_{stag}^2)} \right]^{5/2} \left[1 + 15(f' S_{pv}) f_{stag} b^{3/2} \sqrt{1 + f_{stag}^2} - \frac{45}{4} (f' S_{pv})^2 b^3 (1 + f_{stag}^2)^2 + \dots \right]. \quad (D5)$$

Using Eq. (D5) and following the same steps that lead to Eqs. (12) and (14), the RKE for the present case of the two-piston model with swing can be found, and the result is noted here for completeness

$$(1 + f_{stag}^2) \frac{RKE}{c \cdot KE} \approx f_{stag}^2 - 6(f' S_{pv}) f_{stag} b^{3/2} \sqrt{1 + f_{stag}^2} + \frac{9}{2} (f' S_{pv})^2 b^3 (1 + f_{stag}^2)^2 + \dots \quad (D6)$$

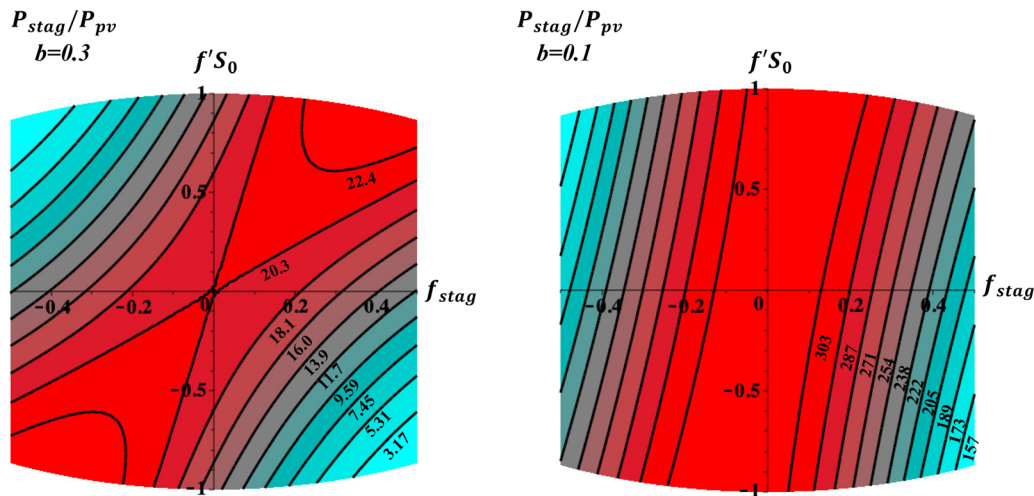


FIG. 14. The degradation of stagnation pressure is shown as a function of mode-1 asymmetry and mode-1 asymmetry swing. For low stagnation pressure implosions, as measured by the parameter $b > 0.1$, the impact of swing can be comparable to the impact of shell asymmetry at stagnation (left frame). For high stagnation pressure implosions ($b < 0.1$) the impact of swing is negligible comparable to the impact of shell asymmetry at stagnation (right frame). The plot range is constrained by $f_{stag}^2 + [f'(S_{pv} - S_{min})]^2 < 1$ so as to guarantee pistons masses remain positive as the asymmetry evolves. The values of the contour levels are labeled on the plot.

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